Trustworthy Collective Reasoning and Evaluation for AI Agents

Approved for Public Release

AI agents built on large language models increasingly evaluate one another and decide together: they judge answers, review papers, supply the rewards that train new models, and vote or discuss. Yet a vote counts a mistake that several agents share as independent evidence, and dishonest agents can collude. AI judges rule on text alone, even when one database query would settle a dispute, and agents trained to agree with their group can learn to agree on mistakes. If five agents tell a Navy watch officer whether a cargo vessel has left port, and three say yes because two read the same outdated schedule and one is compromised, an officer who trusts the vote sends a patrol after a ship still at berth.

The objective is to let a group of AI agents of unknown and correlated reliability, some possibly dishonest, reach trustworthy decisions and evaluations through debate. In a debate, two agents argue for opposite answers until the dispute narrows to a step an impartial judge can verify. The first research goal (RG1) weighs answers with a statistical reliability model, fitted without an answer key, in which a group of agents sharing a mistake counts at most once. It sends a question to debate only when debate is worth its cost and chooses the debaters and the judge impartially, so no agent's ratings of others can get it chosen. RG2 settles disputes in an agentic debate harness: debaters back their steps with tool calls, each side disputes a step of the other's argument, and the judge re-runs the calls and rules on the disputed steps. Planted cases with known answers and random human audits catch colluders. RG3 trains the agents by reinforcement learning on the group's decisions, without human labels. Each agent's answer earns credit for agreeing with the decision when a re-run tool settled it, and otherwise with what the other groups decide, so echoing one's own group earns nothing. Debate messages earn credit for moving the judge, but only in debates the judge settled by re-running tool evidence. While most decisions on a topic are right, wrong ones slow this training without reversing it, and audits stop it on topics where they become frequent.

We will test the methods on three benchmarks with known answers (math and code, multi-document questions with maritime scenarios, and scientific claims and peer review, with answers from reproducibility reports or mistakes we insert) that vary how widely mistakes are shared and how many agents are dishonest. A red team trains agents to collude, and the baselines include a single AI judge and majority vote.

If successful, the project will deliver these methods with their theory (identifiability without an answer key, the price of impartiality, bounds on colluders' gains and harm, and rewards whose expected update follows accuracy), an open debate harness, three benchmarks, and red-team protocols. It will show when a debate among AI agents can be trusted and how few audits keep agents honest. Because people audit only a small share of decisions and training needs no human labels, the methods can scale to AI whose answers people cannot easily evaluate.

For the Department of War, the methods serve staff who need one decision from several AI tools whose reports may be copied, outdated, forged, or compromised, as in maritime domain awareness, intelligence analysis, decision support, evaluating AI before fielding, and unmanned systems sharing reports under limited communications. The aim is fewer missed and false calls, such as whether a vessel has left port, at a fixed budget of human audits, with each decision explained. The 2026 ONR Science and Technology Strategy will explore validating AI results by independent AI critique, and this project supplies methods for such critique when the critics may share mistakes or collude. Navy laboratories could evaluate the open harness on Navy data.